

Population : The group of individuals under study is called Population. It may be finite or infinite.

Sampling :- A part selected from the population is called a sample. The process of selection of a sample is called Sampling.

Parameters and Statistics : The statistical constants of the population such as mean (μ), standard deviation (σ) are called Parameters.

The mean ($\bar{x}$) and standard deviation (s) of a sample are known as statistics.

AIM OF A SAMPLE : The population parameters are generally not known. Then sample statistics are used to approximately determine Population parameters.

Sampling Distribution of a Statistics.

Standard Error : It is the standard deviation of the sampling distribution

Testing A Hypothesis : On the basis of sample information, we make certain decisions about the populations. In taking such decisions we make certain assumptions. These assumptions are known as Statistical Hypotheses. These hypothesis are tested. Assuming the hypothesis is correct we calculate the probability of getting the observed sample. If this probability is less than a certain assigned value, the hypothesis is to be rejected.

H

NULL HYPOTHESIS : It is the assumption that there is no significant difference between the sample statistics and population parameter.

$$H_0: \mu = \mu_0$$

 μ = Population Mean

$$H_1: \mu \neq \mu_0 \quad (\mu > \mu_0 \text{ or } \mu < \mu_0)$$

 μ_0 = Assumed Mean (Sample Mean)

$$H_1: \mu > \mu_0$$

 $\rightarrow$ RIGHT-TAILED TEST

$$H_1: \mu < \mu_0$$

 $\rightarrow$ LEFT-TAILED TEST

CRITICAL VALUE : Let the sample statistics 't' lie in a certain region R. If we decide that there is difference between

sample statistics and population parameter
i.e. H_0 is Rejected. then Region R is
called critical region. The complementary
region is called region of acceptance.

Level of significance: (α) [L.O.S]

α is the probability that a statistic
lies in region R .

$$P\left(t \in \frac{R}{H_0}\right) = \alpha$$

or

$$P\left(t \in \frac{\bar{R}}{H_1}\right) = \beta$$

Where t = statistic of sample
 R = critical region

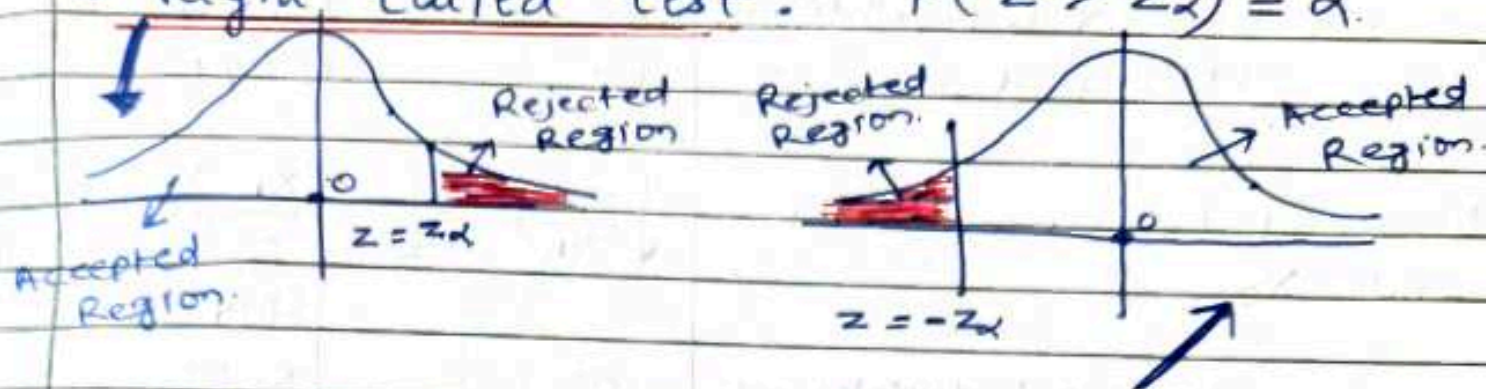
$\bar{R}$ = region of
acceptance.

Critical Value: It is the value of
test statistics which separates the
region of rejection and region of
acceptance & denoted by Z_α .

→ NOTE For large samples ($n \geq 30$)
the test statistic follow a Normal
Distribution $Z = \frac{t - E(t)}{S.E(t)} \sim N(0,1)$

→ central limit Theorem

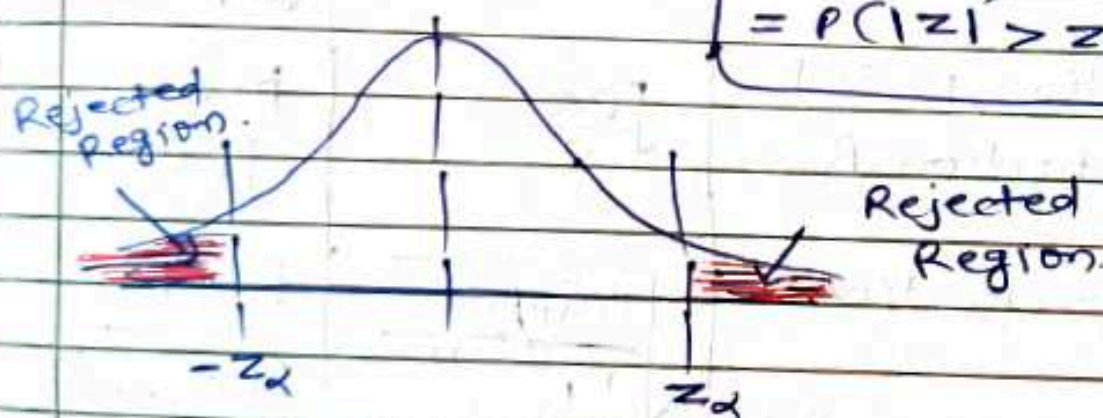
• Right tailed test: $P(Z > Z_\alpha) = \alpha$.



• Left tailed test: $P(Z < -Z_\alpha) = \alpha$

• Two tailed test:

$$P(Z > Z_\alpha) + P(Z < -Z_\alpha) = P(|Z| > Z_\alpha) = \alpha$$



TEST	C.V.	Level of Significance		
		1%	2%	5%
Two Tailed	$ Z_\alpha $	2.58	2.33	1.96
Right Tailed	Z_α	2.33	2.055	1.645
Left Tailed	$-Z_\alpha$	-2.33	-2.055	-1.645

Confidence Interval: It is the interval in which population parameters lie & its end points are called Confidence Limits.

TEST OF SIGNIFICANCE OF DIFFERENCE BETWEEN	S.E.	Statistic Z
① Sample Mean $\bar{X}$ & population Mean μ	$\frac{\sigma}{\sqrt{n}}$	$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$
② Mean of two samples $\bar{X}_1$ and $\bar{X}_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	$\frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$
③ Sample standard deviation = S Population S.D = σ	$\frac{\sigma}{\sqrt{2n}}$	$\frac{S - \sigma}{\sigma/\sqrt{2n}}$
④ Samples standard deviation S_1 & S_2	$\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$	$\frac{S_1 - S_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}}$
⑤ Sample proportion p and population proportion P	$\sqrt{\frac{PQ}{n}}$	$\frac{p - P}{\sqrt{\frac{PQ}{n}}}$
⑥ Two sample proportions p_1 & p_2	$\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$	$\frac{p_1 - p_2}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}}$

Errors:

TYPE I Error: If H_0 is rejected while it should have been accepted.

TYPE II Error: If H_0 is accepted while it should have been accepted.

Confidence Limits :- $\mu - 1.96\sigma, \mu + 1.96\sigma$
 $\mu - 2.58\sigma$

→ Confidence Limits : At
 $\mu - 1.96\sigma, \mu + 1.96\sigma$ 95% confidence Interval
 $\mu - 2.58\sigma, \mu + 2.58\sigma$ 99% " "

→ Confidence Intervals
 $(\mu - 1.96\sigma, \mu + 1.96\sigma)$ 95%
 $(\mu - 2.58\sigma, \mu + 2.58\sigma)$ 99%

→ TEST OF SIGNIFICANCE
(Large samples : $N > 30$)

Test statistics = $Z = \frac{x - \mu}{\sigma}$

(i) If $|Z| < 1.96 \Rightarrow H_0$ is accepted at 5% l.o.s

(ii) If $|Z| < 2.58 \Rightarrow H_0$ is accepted at 1% l.o.s

Q A die was thrown 9000 times and 3240 times 5 or 6 turned up. Does this data indicate that the die is unbiased?

Solⁿ $P(5 \text{ or } 6) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3} = p$

$$Q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}, \quad n = 9000$$

$$n = 9000, \quad \text{Observed proportion of success} \\ = \hat{p} = \frac{3240}{9000} = 0.36$$

$$\text{Test statistic: } z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} = \frac{0.36 - 0.33}{\sqrt{\frac{1}{3} \times \frac{2}{3} \times \frac{1}{9000}}}$$

$$z = 5.43$$

$$|z| > 1.96 \quad \text{at } 5\% \text{ l.o.s.}$$

$\therefore H_0$ is rejected

H_0 : Die is unbiased

H_1 : Die is not unbiased ($\therefore$ Two Tailed Test)

Q In a big city two samples of persons are drawn. In one sample of size 100 the average income of persons is Rs 210 and S.D is 10 and in other samples of size 150 persons the average income is Rs 220 & S.D is 12. Also given is that the S.D. of income

of people of city is Rs 11. Test if there is any significant difference between the two average incomes.

Soln H_0 : there is no difference in sample means : $\bar{X}_1 = \bar{X}_2$

H_1 : $\bar{X}_1 \neq \bar{X}_2$ ($\therefore$ Two Tailed Test)

$$n_1 = 100 \quad n_2 = 150 \quad S_1 = 10 \quad \sigma = 11$$

$$\bar{X}_1 = 210 \quad \bar{X}_2 = 220 \quad S_2 = 12$$

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{210 - 220}{\sqrt{\frac{10^2}{100} + \frac{12^2}{150}}} = -7.1428$$

$$|Z| = 7.1428 > 1.96$$

$\therefore H_0$ is rejected

$\therefore$ Two Sample Means differ.

SMALL SAMPLES ($N < 30$)

Degree of freedom :- D.O.f : It is the total number of observations minus number of restrictions.

eg: If we want to choose four whose sum is 40 then 3 nos can be chosen independently but 4th number = $40 - (\text{sum of three nos})$

$$\therefore \text{D.O.f} = 4 - 1 = 3$$

D.O.f is denoted by ν

t-test For the Mean of Random

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t-test

$$t = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Where $\bar{x}$ is Sample Mean
 μ " Population Mean

$$\sigma^2 = \frac{n}{n-1} s^2$$

s = S.D. of Sample

Q A sample of 20 items has mean 42 units & S.D. 5 units. Test the hypothesis that it is a random sample from a normal population with mean 45 units.

$$n=20, \bar{X}=42, s=5, \text{d.o.f.} = 19$$

H_0 : Sample Mean = Population Mean.

H_1 : Sample Mean $\neq$ Population Mean

($\therefore$ Two Tailed Test).

$$\sigma^2 = \frac{20}{19} \times 5^2 = 26.31$$

$$t = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{42 - 45}{5.129 / \sqrt{20}} = -2.615$$

$$|t| = 2.615 > t_{0.05}$$

$$(t_{0.05} = 2.09 \text{ for } 19 \text{ d.o.f.})$$

$\therefore H_0$ is rejected.

Teacher's Signature: _____

t-test for difference of Means
of Two small samples:

$$t = \frac{\bar{x} - \bar{y}}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \bar{x} = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i, \quad \bar{y} = \frac{1}{n_2} \sum_{i=1}^{n_2} y_i$$

$$\sigma^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{i=1}^{n_2} (y_i - \bar{y})^2 \right]$$

$$D.O.f = n_1 + n_2 - 2$$

This t is to be compared with table value of t at 5% l.o.s.

Q In a school the heights of six-randomly chosen girls are 63, 65, 68, 69, 71, 72 inches and those of 9 randomly chosen boys are 61, 62, 65, 66, 69, 70, 71, 72, 73 inches.

Test if the girls are taller than boys.

Let $\bar{x}_1$ = sample heights of girls $n_1 = 6$
 $\bar{x}_2$ = " " " " boys. $n_2 = 9$.

H₀: Mean of first sample = Mean of II sample.

$$\bar{x}_1 = \frac{\sum x_i}{n_1} = 68 \quad \bar{x}_2 = \frac{\sum x_i}{n_2} = 67.66$$

$$\sum (x_1 - \bar{x}_1)^2 = 60$$

$$\sum (x_2 - \bar{x}_2)^2 = 152.00$$

$$\sigma^2 = \frac{60 + 152}{6 + 9 - 2} = 16.3077 \Rightarrow \sigma = 4.038$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{68 - 67.66}{4.038 \sqrt{\frac{1}{6} + \frac{1}{9}}} = 0.3031$$

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$$D.O.F = n_1 + n_2 - 2 = 13$$

table value of $t = 1.77$ at 13 d.o.f
at 5% l.o.s.

$$\therefore 0.3031 < 1.77$$

$\therefore H_0$ accepted.

Chi-Square Variate and Test For Population Variance :-

The square of a standard normal variate is known as Chi-square variate (χ^2) with one degree of freedom.

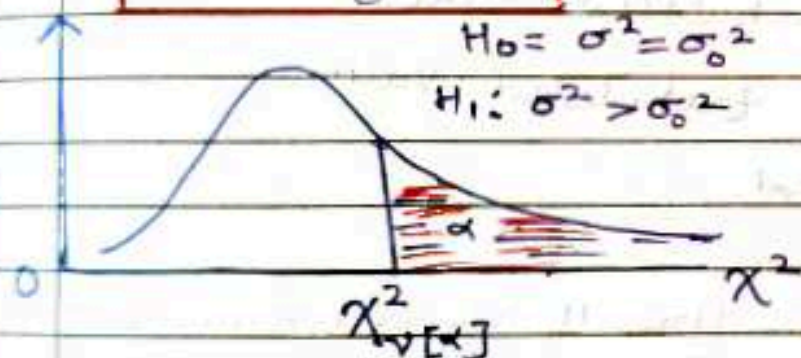
$Z = \frac{x - \mu}{\sigma}$ is a normal variate

$\Rightarrow \left(\frac{x - \mu}{\sigma}\right)^2$ is a Chi-square Variate

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

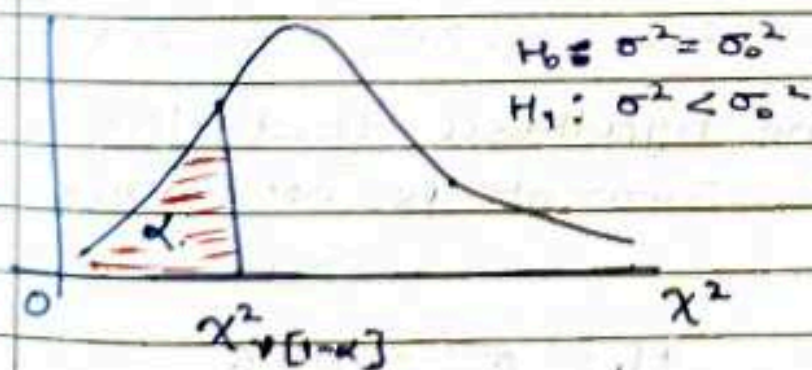
s^2 = sample variance

σ^2 = Population variance.



$\chi^2_{v(\alpha)}$ = value of chi-square variate for v d.f.

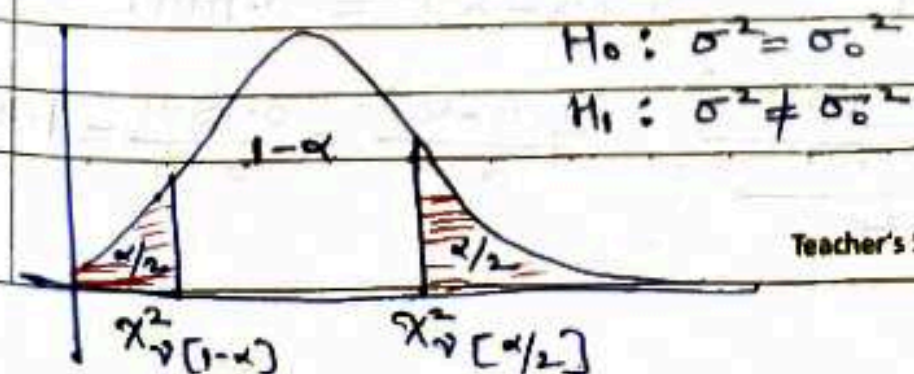
$$P[\chi^2 > \chi^2_{v(\alpha)}] = \alpha$$



$$P[\chi^2 < \chi^2_{v(1-\alpha)}]$$

NOTE:

In all three cases d.o.f = $v = n - 1$



Teacher's Signature : _____

Q A manufacturer of car batteries claims that the life of his batteries is approximately normally distributed with a S.D. of 0.9 years. If a random sample of 10 of these batteries has a S.D. of 1.2 years, do you think $\sigma > 0.9$ years at $\alpha = 0.05$?

Solⁿ $H_0: \sigma^2 = 0.81$ $H_1: \sigma^2 > 0.81$
 $n = 10$, $\sigma^2 = 0.81$, $s^2 = (1.2)^2 = 1.44$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{9(1.44)}{0.81} = 16 \text{ (calculated value)}$$

It follows χ^2 distribution with d.o.f = $9 = 10 - 1$

$$\chi^2_9(0.05) = 16.92 \text{ (tabulated value)}$$

$\therefore$ calculated value < tabulated value

$\therefore H_0$ is accepted.

Q Following data gives the 11 measurements of the same object on the same instruments:

2.5, 2.3, 2.4, 2.5, 2.7, 2.5, 2.6, 2.6, 2.7, 2.5, 2.3

At 1% level, test the hypothesis that the variance of the instrument is not more than 0.16.

Solⁿ $H_0: \sigma^2 \leq 0.16$

$H_1: \sigma^2 > 0.16$

$\bar{x} = \frac{27.6}{11} = 2.51$

$\sum (x - \bar{x})^2 = 0.1891$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(n-1) \frac{\sum (x - \bar{x})^2}{n-1}}{\sigma^2} = \frac{\sum (x - \bar{x})^2}{\sigma^2} = \frac{0.1891}{0.16} = 1.182$$

$$d.o.f = \nu = n-1 = 11-1 = 10$$

$$\chi^2_{10[0.1]} = 23.2 \quad \therefore \text{calculated value} < \text{tabulated value}$$

$\therefore H_0$ accepted.

CHI-SQUARE TEST AS TEST OF GOODNESS OF FIT

$$\chi^2 = \sum_{i=1}^n \left[\frac{(O_i - E_i)^2}{E_i} \right] \quad \begin{array}{l} O_i = \text{observed frequencies} \\ E_i = \text{expected frequencies} \end{array}$$

$$N = \sum_{i=1}^n O_i = \sum_{i=1}^n E_i$$

NOTE: $d.o.f = \nu = n-1$

Q A die is thrown 276 times and results of these throws are tabulated below:-

Number on the die	1	2	3	4	5	6
Frequency	40	32	29	59	57	59

Test if the die is unbiased using χ^2 -test.

Solⁿ H_0 : Die is unbiased.

Expected frequency = $\frac{\sum x_i}{n} = \frac{276}{6} = 46$

O_i	40	32	29	59	57	59
E_i	46	46	46	46	46	46
$(O_i - E_i)^2$	36	196	289	169	121	169
$\frac{1}{E_i} (O_i - E_i)^2$	0.782	4.261	6.282	3.674	2.63	3.674

$$\chi^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i} = 21.3$$

$\chi^2_5(0.05) = 11.07$ $\therefore$ calculated value > tabulated value

$\therefore H_0$ Rejected.

Regression:- The Regression analysis is a statistical device which help to estimate (or predict) the unknown value of one variable from the known value of another variable.

Line of Regression:- The dots of scatter diagram generally tends to cluster about a well defined direction $\Rightarrow$ there exist a linear relationship between the variables x & y . Such a line of best fit is called line of Regression.



It has two Types:-

(a) Line of regression of y on x :-

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

• coefficient of regression of y on $x = b_{yx} = r \frac{\sigma_y}{\sigma_x}$

$$r^2 = b_{yx} \cdot b_{xy}$$

(b) Line of regression of x on y

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

• coefficient of regression of x on y :-

$$b_{xy} = r \cdot \frac{\sigma_x}{\sigma_y}$$

Teacher's Signature _____

Q From the following data, obtain the two lines of regression.

x	6	2	10	4	8	//
y	9	11	5	8	7	

Solⁿ $dx = x - \bar{x}$, $dy = y - \bar{y}$ where $\bar{x}$ & $\bar{y}$ are assumed means for x & y .

x	y	$dx = x - 6$	$dy = y - 8$	dx^2	dy^2	$dx dy$
6	9	1	1	1	1	1
2	11	-4	3	16	9	-12
10	5	4	-3	16	9	-12
4	8	-2	0	4	0	0
8	7	2	-1	4	1	-2
$\Sigma x = 30$	$\Sigma y = 40$	$\Sigma dx = 5$	$\Sigma dy = 5$	$\Sigma dx^2 = 45$	$\Sigma dy^2 = 25$	$\Sigma dx dy = -21$

The line of regression of x on y is

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{30}{5} = 6, \quad \bar{y} = \frac{\Sigma y}{n} = \frac{40}{5} = 8$$

$$b_{xy} = \frac{\Sigma dx dy - \frac{1}{N} \Sigma dx \Sigma dy}{\Sigma dy^2 - \frac{1}{N} (\Sigma dy)^2} = \frac{-21 - \frac{1}{5}(5)(5)}{25 - \frac{1}{5}(5)^2} = -1.3$$

$$\therefore x - 6 = -1.3(y - 8) \Rightarrow x = -1.3y + 16.4$$

$$b_{yx} = \frac{\Sigma dx dy - \frac{1}{N} \Sigma dx \Sigma dy}{\Sigma dx^2 - \frac{1}{N} (\Sigma dx)^2} = \frac{-21 + \frac{1}{5}(5)(5)}{45 - \frac{1}{5}(5)^2} = -0.65$$

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$\therefore$ The line of regression of x on y is
 $y - 8 = -0.65(x - 6) \Rightarrow y = -0.65x + 11.9.$

Analysis Of Variance ANOVA

①

• It is developed by R.A. Fisher in 1920.

Variance: It is defined as the expectation of the squared deviations of a R.V. from its mean. (σ^2 or s^2)

ANOVA: It is used to compare more than two populations or populations having more than two subgroups.

ANOVA = $\frac{\text{Variability between the MEANS}}{\text{variability within the distribution}}$

Total Variance = variability between the Means + variability within the distribution.

Assumptions:

- ① Each population is having normal Distribution
- ② The populations from which the samples are drawn have the equal variance.
i.e. $S_1^2 = S_2^2 = S_3^2 \dots = S_k^2$ for K-samples.
- ③ Each sample is drawn randomly & hence they are independent.

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_n$$

$$H_1: \mu_1 \neq \mu_2 \neq \mu_3 \neq \dots \neq \mu_n$$

(2)

Classification

one factor
One way ANOVA

Two factor
Two way ANOVA

One Way Anova : It is classified according to using one factor / one criteria.

A	B	C
2	3	4
4	5	6
6	7	8
12	15	18

① STATE H_0 & H_1

$$H_0 : \bar{x}_A = \bar{x}_B = \bar{x}_C$$

$$H_1 : \bar{x}_A \neq \bar{x}_B \neq \bar{x}_C$$

② Calculate the variance between the samples :-

Calculate Means of each samples.

$$\bar{x}_A = \frac{12}{3} = 4, \quad \bar{x}_B = \frac{15}{3} = 5, \quad \bar{x}_C = \frac{18}{3} = 6$$

$$\bar{\bar{x}} = \frac{\bar{x}_A + \bar{x}_B + \bar{x}_C}{3} = \frac{4 + 5 + 6}{3} = 5$$

Grand Average of Means

$(\bar{x}_A - \bar{\bar{x}})$	$(\bar{x}_A - \bar{\bar{x}})^2$	$(\bar{x}_B - \bar{\bar{x}})$	$(\bar{x}_B - \bar{\bar{x}})^2$	$(\bar{x}_C - \bar{\bar{x}})$	$(\bar{x}_C - \bar{\bar{x}})^2$
$-1 = 4 - 5$	1	$5 - 5 = 0$	0	$6 - 5 = 1$	1
$-1 = 4 - 5$	1	$5 - 5 = 0$	0	$6 - 5 = 1$	1
$-1 = 4 - 5$	1	$5 - 5 = 0$	0	$6 - 5 = 1$	1
$\Sigma(\bar{x}_A - \bar{\bar{x}})^2$	3	$\Sigma(\bar{x}_B - \bar{\bar{x}})^2$	0	$\Sigma(\bar{x}_C - \bar{\bar{x}})^2$	3

Sum of squares ^{between} ~~within~~ samples = $3 + 0 + 3$
= 6

Good Write

5) Calculate variance within the samples.

$$\bar{x}_A = 4, \quad \bar{x}_B = 5, \quad \bar{x}_C = 6$$

$A - \bar{x}_A$	$(A - \bar{x}_A)^2$	$B - \bar{x}_B$	$(B - \bar{x}_B)^2$	$C - \bar{x}_C$	$(C - \bar{x}_C)^2$
$2-4=-2$	4	$3-5=-2$	4	$4-6=-2$	4
$4-4=0$	0	$5-5=0$	0	$6-6=0$	0
$6-4=2$	4	$7-5=2$	4	$8-6=2$	4
$\Sigma(A-\bar{x}_A)^2$	8	$\Sigma(B-\bar{x}_B)^2$	8	$\Sigma(C-\bar{x}_C)^2$	8

Sum of squares within samples = $8+8+8=24$

Calculate the ratio of

Source of Variation	Sum of Squares	D.O.F	Mean Sum of Squares
Between the samples	$SSC = 6$	$\nu_1 = C-1$ $3-1=2$	$MSC = SSC/C-1 = 3$
Within the samples	$SSE = 24$	$\nu_2 = n-C$ $9-3=6$	$MSE = SSE/n-C$ $= 24/6$ $= 4$

$SSC =$ Sum of squares b/w samples

$SSE =$ Sum of squares within samples

$MSC =$ Mean sum of squares b/w "

$MSE =$ Mean sum " " within "

$C =$ No. of columns

$n =$ No. of observation

F

$F = MSC/MSE$

$F = 3/4 = 0.75$

↓
Calculated value

Tabulated value = F value = 5.14

$\therefore$ Tabulated value > calculated value.

$\therefore H_0$: accepted $\therefore H_0$ is correct.

Teacher's Signature _____

TWO WAY ANOVA :-

- It is classified according to two factors or two criteria.

① Four students A, B, C, D study hours are as follow:

Days	A	B	C	D
Monday	2	3	4	5
Tuesday	4	5	6	7
Wednesday	6	7	8	9

Two way ANOVA can be applied and variance can be determined.

- ① Between the columns (b/w A, B, C, D)
- ② Between the rows (b/w Mon, Tue, Wed)

($\therefore$ Here we are finding variance for two criteria $\therefore$ TWO WAY ANOVA is Applied)

Source of variation	Sum of square	D.O.f	Mean Sum of square	Ratio of F
Between the columns	$S.S.C = 15$	$V = C - 1 = 4 - 1 = 3$	$MSC = \frac{SSC}{C-1} = \frac{15}{3} = 5$	$\frac{MSC}{MSE} = \frac{5}{0} = \infty$
Between rows	$S.S.R = 32$	$V = R - 1 = 3 - 1 = 2$	$MSR = \frac{SSR}{R-1} = \frac{32}{2} = 16$	$\frac{MSR}{MSE} = \frac{16}{0} = \infty$
Residual or Error	$SSE = 0$	$V = (n-1) - (C-1) - (R-1) = (12-1) - 3 - 2 = 6$	$MSE = \frac{SSE}{n-1-C-1} = \frac{0}{6} = 0$	
	$BST = 47$	$V = n - 1 = 12 - 1 = 11$		

① Calculation of Grand total & correction factor.

Day	A	B	C	D	Total
M	2-5=-3	3-5=-2	4-5=-1	5-5=0	-6
T	4-5=-1	5-5=0	6-5=1	7-5=2	2
W	6-5=1	7-5=2	8-5=3	9-5=4	10
Total	-3	0	3	6	6=T

↓
Grand Total (T)

$$\text{Correction factor} = \frac{T^2}{N}$$

$$= \frac{6^2}{12} = 3$$

Where N = Total No.

of observation.

SSC = Sum of squares b/w columns.

$$SSC = \frac{A^2}{N_A} + \frac{B^2}{N_B} + \frac{C^2}{N_C} + \frac{D^2}{N_D} - \frac{T^2}{N} = \left\{ \frac{9}{3} + \frac{0}{3} + \frac{9}{3} + \frac{36}{3} \right\} - \frac{36}{12} = (3+3+12) - 3 = 18 - 3 = 15$$

$$SSR = \frac{M^2}{N_M} + \frac{T^2}{N_T} + \frac{W^2}{N_W} - \frac{T^2}{N} = \frac{36}{4} + \frac{4}{4} + \frac{100}{4} - \frac{36}{12} = 32$$

$$SST = \text{Total Sum of squares} = ((-3)^2 + (-1)^2 + (1)^2 + (-2)^2 + 0^2 + 2^2 + (-1)^2 + 1^2 + 3^2 + 0^2 + 2^2 + 4^2) - \frac{T^2}{N}$$

$$SST = 47$$

SSE = Total Sum of squares due to error.

$$= SST - (SSC + SSR) = 47 - (15 + 32) = 0$$

F value for $\gamma_1 = 6, \gamma_2 = 3$ $F_{0.05} = 4.76$ } Tabulated value

F value for $\gamma_1 = 6, \gamma_2 = 2$ $F_{0.05} = 5.34$ }

cal: Ratio of F = calculated value of F = $\infty >$ Tabulated value (4.76)

Reco: calculated value of F = $\infty >$ Tabulated value (5.34)

∴ Yes there is significant difference

Teacher's Signature _____

among days and student w.r.t. Study hours.